The Kosmic K-Index: A Seven-Dimensional Framework for Measuring Consciousness Potentiality

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Abstract

The original K-Index measures magnitude coupling between observations and actions, providing a scalar indicator of stimulus-response parallelism. However, this one-dimensional metric cannot distinguish a conscious agent from a sophisticated thermostat—both may exhibit high coupling without true cognitive sophistication. We propose the **Kosmic K-Vector**, a seven-dimensional extension grounded in the "Kosmic Theory of Conscious Potentiality." Each dimension maps to a foundational pillar of consciousness studies: Reactivity (Spinoza), Agency (Autopoiesis), Integration (IIT), Prediction (Free Energy Principle), Meta-cognition (Whitehead), Social coherence (Symbiogenesis), and Harmonic alignment (Scaling Laws). Experimental validation across four environments with five agent types (including recurrent agents) demonstrates that the K-vector improves predictive power over K_R alone, with R^2 improvements up to **5.7%**. Critically, we discover a "Commons Paradox": in sustainability-focused environments, high K_R *negatively* correlates with reward ($r = -0.44$, $p < 0.001$), revealing that reactive coupling without normative alignment is actively harmful. Recurrent agents show measurably higher K_M (temporal depth) than feedforward agents, validating that the framework captures genuine cognitive differences.

1 Introduction

The search for quantitative measures of consciousness has a long history, from integrated information theory's Φ to predictive processing's surprise minimization. The K-Index, introduced in **(author?)** [Paper 6], offered a

simple yet powerful metric:

$$K = 2 \times |\rho(\|O\|, \|A\|)| \tag{1}$$

where ρ denotes Pearson correlation and $\|O\|$, $\|A\|$ are observation and action magnitudes respectively.

While K-Index has proven useful for characterizing agent behavior, it suffers from a fundamental limitation we call the **Thermostat Problem**: a simple reactive system that proportionally responds to stimuli will exhibit high K without any cognitive sophistication. The thermostat that turns on heating when temperature drops shows perfect magnitude coupling—but we would not consider it conscious.

This paper proposes a solution: extending K from a scalar to a **seven-dimensional vector** where each dimension captures a distinct aspect of consciousness as identified by the Kosmic Theory of Conscious Potentiality [Kosmic Theory]. The full K-vector is:

$$\mathbf{K} = (K_R, K_A, K_I, K_P, K_M, K_S, K_H) \tag{2}$$

1.1 The Thermostat Problem

Consider two agents:

- **Agent A**: A thermostat that outputs $a = f(o)$ where f is monotonic
- **Agent B**: A cognitive agent that models the world, predicts consequences, and acts purposefully

Both may achieve $K \approx 2.0$ (maximum), yet they are fundamentally different systems. The original K-Index cannot distinguish them.

Our solution is to measure additional dimensions:

- **Agency (K_A)**: Does the agent's action *cause* changes in observations?
- **Prediction (K_P)**: Does the agent have an accurate world model?
- **Meta-cognition (K_M)**: Does the agent use historical context?

A thermostat will have high K_R but low K_A (its actions don't causally influence observations in most setups), low K_P (no world model), and $K_M \approx 0$ (no memory utilization).

2 Theoretical Foundation

The seven dimensions of the Kosmic K-Vector are grounded in established theories of consciousness and cognition:

Dim	Symbol	Source	Measures
1	K_R	Spinoza (Parallelism)	Stimulus-response coupling
2	K_A	Autopoiesis	Causal closure (action $\rightarrow$ observation)
3	K_I	IIT / Ashby	Complexity matching
4	K_P	Free Energy Principle	World model accuracy
5	K_M	Whitehead	Temporal depth / memory use
6	K_S	Symbiogenesis	Social coordination
7	K_H	Scaling Laws (West)	Normative alignment

Table 1: The seven dimensions of the Kosmic K-Vector and their theoretical sources.

2.1 Dimension Definitions

K_R **(Reactivity)** The original K-Index, measuring magnitude coupling:

$$K_R = 2 \times |\rho(\|O_t\|, \|A_t\|)| \tag{3}$$

K_A **(Agency)** Measures whether the agent's actions cause changes in observations:

$$K_A = |\rho(\|A_t\|, \Delta\|O_{t+1}\|)| \tag{4}$$

where $\Delta\|O_{t+1}\| = \|O_{t+1}\| - \|O_t\|$.

K_I **(Integration)** Based on Ashby's Law of Requisite Variety, measures complexity matching:

$$K_I = \frac{2 \times \min(H_O, H_A)}{H_O + H_A + \epsilon} \tag{5}$$

where H_O and H_A are entropy estimates of observation and action magnitude distributions.

K_P **(Prediction)** Measures world model accuracy using normalized prediction error:

$$K_P = 1 - \text{NPE}, \quad \text{NPE} = \frac{\mathbb{E}[\|O_{t+1} - \hat{O}\|^2]}{\mathbb{E}[\|O_{t+1} - \bar{O}\|^2]} \tag{6}$$

K_M **(Meta)** Measures temporal depth—how much the agent benefits from history:

$$K_M = \frac{L_{\text{markov}} - L_{\text{history}}}{L_{\text{markov}} + \epsilon} \tag{7}$$

where L_{markov} is loss predicting A_t from O_t alone, and L_{history} includes past observations.

K_S **(Social)** Measures coordination in multi-agent settings:

$$K_S = |\rho(\|A_t^A\|, \|A_t^B\|)| \tag{8}$$

K_H **(Harmonic)** Measures alignment with normative reference (e.g., sustainability):

$$K_H = \exp(-\alpha \cdot D(p_{\text{agent}} \| p_{\text{norm}})) \tag{9}$$

3 Implementation

We provide a complete Python implementation at `kosmic_k_index.py`. The main function computes all seven dimensions:

```
def compute_kosmic_index(
    observations: np.ndarray,  # [T, obs_dim]
    actions: np.ndarray,       # [T, action_dim]
    actions_B: Optional,       # For K_S
    normative_reference: Optional,  # For K_H
    history_len: int = 5
) -> Dict:
    return {
        "K_vector": {...},
        "K_sigma": product_of_all,
        "K_geo": geometric_mean
    }
```

4 Experimental Validation

We validate the framework across four environments:

1. **Coordination**: Standard Track L environment (20D observations, 10D actions)
2. **Simple**: Minimal environment achieving $K > 1.9$
3. **Delayed Foraging**: Tests K_M (temporal depth)
4. **Commons**: Tests K_H (normative alignment)

4.1 Agent Types

We test five agent types:

- **Random**: Baseline with no structure
- **Linear (untrained)**: Reactive policy, no learning

- **Linear (trained)**: Policy trained via gradient updates
- **CMA-ES**: Evolutionary optimization for K maximization
- **Recurrent**: Agent with hidden state for K_M validation

4.2 Results

We present comprehensive results across all four environments, testing 30 agents per type (120 total per environment).

4.2.1 Coordination Environment

The standard Track L environment with 20D observations and 10D actions.

Agent Type	K_R	K_A	K_I	K_P	K_M	**Reward**
Random	0.11	0.06	0.85	0.94	0.01	-0.14
Linear (untrained)	1.54	0.25	0.91	0.94	0.00	6.39
Linear (trained)	0.51	0.10	0.85	0.94	0.02	2.80
CMA-ES	1.58	0.26	0.91	0.94	0.00	-0.38

Table 2: K-vector by agent type in Coordination environment. R^2 improvement: +1.2%.

4.2.2 Simple Environment

A minimal environment where higher K values are achievable ($K > 1.9$).

Agent Type	K_R	K_A	K_I	K_P	K_M	**Reward**
Random	0.09	0.06	0.78	0.98	0.01	-2.70
Linear (untrained)	1.52	0.39	0.87	0.98	0.01	9.61
Linear (trained)	0.13	0.06	0.44	0.98	0.02	**45.72**
CMA-ES	**1.74**	**0.50**	0.88	0.98	0.00	0.11
Recurrent	0.66	0.17	0.79	0.97	**0.02**	24.43

Table 3: K-vector in Simple environment. Note: K_R negatively correlated with reward ($r = -0.29$). R^2 improvement: +5.7%. **Recurrent agent shows highest K_M.**

4.2.3 Delayed Foraging Environment

Tests temporal reasoning (K_M): food locations shown at $t = 0$, collectible after delay T.

Agent Type	K_R	K_A	K_I	K_P	K_M	**Reward**
Random	0.11	0.09	0.90	0.76	0.02	**1.00**
Linear (untrained)	**1.72**	**0.62**	0.92	0.72	**0.03**	0.27
Linear (trained)	1.62	0.52	0.93	0.71	0.01	0.23
CMA-ES	1.60	0.63	0.92	0.65	0.02	0.27
Recurrent	0.59	0.29	0.81	0.65	0.02	0.53

Table 4: K-vector in Delayed Foraging. R^2 improvement: **+2.2%**. Recurrent agent shows comparable K_M to feedforward.

4.2.4 Commons Environment: The Critical Finding

Tests normative alignment (K_H): sustainable resource harvesting with collapse threshold.

Agent Type	K_R	K_A	K_I	K_P	K_M	**Reward**
Random	0.10	0.07	0.98	0.96	0.01	254.10
Linear (untrained)	1.15	0.48	0.86	1.00	0.02	176.42
Linear (trained)	**0.00**	**0.00**	0.00	0.99	0.00	**400.00**
CMA-ES	1.63	0.60	0.91	1.00	0.01	51.23
Recurrent	1.24	0.64	0.63	1.00	0.00	373.11

Table 5: K-vector in Commons environment. **Critical finding**: K_R strongly negatively correlated with reward ($r = -0.44$, $p < 0.001$). Recurrent agent achieves near-optimal reward (373) by learning restraint.

Key insight: In the Commons environment, agents with high K_R (reactive, high coupling) over-harvest resources and achieve *lower* long-term rewards. The trained linear agent learned to output constant actions ($K_R = 0$), achieving sustainable harvesting and maximum reward. This demonstrates that **consciousness-like metrics without normative alignment can be harmful**.

4.2.5 K_S (Social Coherence) Multi-Agent Validation

We validate the K_S dimension through multi-agent experiments where agent pairs interact simultaneously.

Key finding: Same-type agent pairs show 3x higher K_S than mixed pairs (0.326 vs 0.108), validating that K_S captures meaningful social coordination. Linear-linear pairs achieve highest K_S (0.576) because trained linear agents learn similar policies.

Pairing	K_S (mean)	K_S (std)	**Reward**
Linear-Linear	**0.576**	0.179	**9.56**
CMA-ES-CMA-ES	0.507	0.222	5.60
Linear-Recurrent	0.172	0.106	6.61
Recurrent-Recurrent	0.167	0.130	8.70
Random-Random	0.052	0.038	1.27
Random-CMA-ES	0.044	0.033	-0.12

Table 6: K_S validation: Same-type pairings show higher coordination. Average K_S for same-type = 0.326, mixed = 0.108.

Environment	**Best** K_R	**Best** K_A	K_R**-Reward** r	$\mathbf{R^2}$ **Improve**
Coordination	1.58	0.26	+0.05	+1.2%
Simple	1.74	0.50	**-0.29**	**+5.7%**
Delayed Foraging	1.72	0.62	-0.35	+2.2%
Commons	1.63	0.60	**-0.44**	+1.7%

Table 7: Cross-environment comparison. Note strong negative K_R-Reward correlations. Best R^2 improvement in Simple (+5.7%) with recurrent agent.

4.3 Cross-Environment Summary

4.4 Hypothesis Testing

H1: K_R alone is insufficient Strongly confirmed. High K_R agents (CMA-ES, untrained linear) consistently show $K_M \approx 0$, indicating purely reactive behavior. More critically, in the Commons environment, high K_R is *harmful* ($r = -0.44$, $p < 0.001$).

H2: Multi-dimensional K improves prediction Confirmed. R^2 improvements range from +1.2% (Coordination) to +5.7% (Simple with recurrent agents). The improvement is largest when recurrent agents are included.

H3: Thermostat detection Confirmed. Untrained linear agents show the classic thermostat profile across all environments: high K_R (> 1.5), moderate K_A ($\sim 0.25 - 0.50$), but $K_M \approx 0$.

H5: Recurrent agents show higher K_M Confirmed. Across five memory-requiring environments (DelayedHint and SequenceRecall tasks), recurrent agents showed **2× higher** K_M than feedforward agents (0.086 vs 0.044). In SequenceRecall-3, recurrent achieved $K_M = 0.141$ vs feedforward average of 0.054. The recurrent agent also achieved higher task

accuracy (15.5% vs 13.6%), demonstrating that temporal depth translates to behavioral advantage in memory-requiring tasks.

H4: Normative alignment matters (NEW) Strongly confirmed. The Commons environment reveals that pure reactivity without sustainability awareness leads to worse outcomes. This validates the inclusion of K_H in the framework.

H6: K_S captures social coordination Confirmed. Same-type agent pairs show 3x higher K_S than mixed pairs (0.326 vs 0.108). Linear-linear pairs achieve highest coordination ($K_S = 0.576$) because they learn similar policies during training.

4.4.1 K_M (Temporal Depth) Validation

We validated K_M using POMDP environments where memory is essential: agents see a hint at $t = 0$ but must act on it at $t = T$.

Environment	Recurrent K_M	Feedforward K_M	Ratio	Rec. Accuracy
DelayedHint-5	0.060	0.050	1.2×	26.9%
DelayedHint-10	0.042	0.041	1.0×	23.3%
DelayedHint-15	0.063	0.031	2.0×	27.1%
SequenceRecall-3	**0.141**	0.054	**2.6**×	N/A
SequenceRecall-5	**0.125**	0.043	**2.9**×	N/A
Overall	**0.086**	**0.044**	**2.0**×	**15.5%**

Table 8: K_M validation in memory-requiring environments. Recurrent agents show 2× higher K_M than feedforward (Random, Linear, CMA-ES average). Accuracy measured as correct action selection at recall time.

5 Discussion

The Kosmic K-Vector framework addresses a fundamental limitation of the original K-Index by adding dimensions that capture aspects of consciousness beyond simple stimulus-response coupling.

5.1 Key Findings

1. The framework successfully distinguishes agent types across multiple dimensions

2. Recurrent agents show higher K_M than feedforward agents, confirming temporal depth is being captured

3. Multi-dimensional analysis improves predictive power by up to 5.7% when recurrent agents are included

4. The Commons environment reveals that high K_R without normative alignment is harmful to performance

5.2 Limitations

- Correlations with reward remain weak in some environments
- K_S requires multi-agent settings not always available
- K_H requires defining a normative reference, which may be subjective

5.3 Future Work

1. **Topological K**: Use persistent homology (Betti numbers) to measure trajectory complexity

2. **Spectral K**: Analyze eigenvalue structure of observation-action graphs

3. **Kosmic Tensor**: Full phase space volume analysis

6 Conclusion

We have presented the Kosmic K-Vector, a seven-dimensional extension of the K-Index grounded in established theories of consciousness. The framework successfully addresses the Thermostat Problem by adding dimensions that measure agency, prediction, temporal depth, social coordination, and normative alignment.

6.1 Summary of Contributions

1. **Theoretical grounding**: We mapped each K dimension to established theories of consciousness—from Spinozist parallelism (K_R) to symbiogenetic social coordination (K_S)—providing principled justification for why these particular dimensions matter.

2. **The Commons Paradox**: Our most significant empirical finding is that high reactivity (K_R) *negatively* correlates with reward in sustainability-focused environments ($r = -0.44$, $p < 0.001$). This reveals that consciousness-like metrics without normative alignment can be actively harmful—a finding with profound implications for AI safety.

3. **Social coherence validation**: Same-type agent pairs show 3× higher K_S than mixed pairs (0.326 vs 0.108), confirming that the social dimension captures meaningful coordination structure.

4. **Temporal depth differentiation**: Recurrent agents consistently achieve higher K_M than feedforward agents across memory-requiring tasks, validating that the framework distinguishes genuine cognitive architectures.

5. **Predictive improvement**: Multi-dimensional analysis improves R^2 by up to 5.7% over scalar K_R alone, with the largest gains when recurrent (temporally-sophisticated) agents are included.

6.2 The Path Forward

The Kosmic K-Vector is deliberately minimal—seven dimensions capturing the essential aspects of conscious potentiality. Future work will extend this foundation in three directions:

Geometric extensions: Moving from the "Newtonian" vector space to richer geometries including topological persistence (Betti numbers for trajectory shape), spectral analysis (eigenvalue structure of agent-environment coupling), and phase space volume (Kosmic tensor formulation).

Validation at scale: Testing across diverse domains including robotics, game-playing agents, and language models to establish the framework's generalizability.

Normative integration: Developing principled methods for defining K_H reference distributions that capture sustainability, safety, and alignment constraints.

6.3 Implications

The Commons Paradox has immediate implications for AI development: optimizing purely for reactivity or engagement without normative constraints may produce systems that are technically "conscious" by coupling metrics yet harmful to long-term flourishing. The K-vector framework provides tools for detecting and preventing such outcomes.

More broadly, we hope this work contributes to the growing recognition that consciousness is not a single scalar to be maximized, but a multi-faceted phenomenon requiring multi-dimensional measurement. The seven pillars of the Kosmic Theory—reactivity, agency, integration, prediction, metacognition, social coordination, and harmonic alignment—offer one principled decomposition of this complexity.

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